

ALMATY INTL, Kazakhstan

WMO: 368720

Lat: 43.352N

Lon: 77.041E

Elev: 2234

StdP: 13.55

Time Zone: 6.00 (E06)

Period: 96-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-7.3	-0.7	-11.1	3.3	-7.0	-4.9	4.6	-0.8	16.4	34.7	13.1	31.7	3.0	190	0.496

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	26.6	95.2	63.9	92.8	63.5	89.6	62.9	67.8	85.0	66.3	84.8	65.1	83.2	7.0	10

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.6	92.1	72.8	60.7	86.1	72.3	58.9	80.6	71.5	33.5	85.7	32.3	84.2	31.2	82.8	79.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
15.9	13.0	11.0	-12.0	99.4	7.8	2.4	-17.6	101.1	-22.1	102.5	-26.5	103.9	-32.2	105.6		
			-12.4	71.2	7.6	3.1	-17.9	73.5	-22.3	75.3	-26.6	77.0	-32.2	79.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	49.7	19.9	25.0	40.3	53.7	62.1	71.0	75.3	72.9	63.4	50.7	36.0	24.6
	DBStd	20.91	10.74	10.98	10.28	7.80	6.42	5.37	4.80	6.04	7.17	8.19	10.05	9.58
	HDD50	3321	934	699	323	52	3	0	0	0	3	88	431	788
	HDD65	6467	1399	1119	766	343	131	15	1	9	113	447	871	1253
	CDD50	3212	0	0	21	162	378	629	786	710	406	110	10	0
	CDD65	881	0	0	0	3	41	193	321	254	65	4	0	0
	CDH74	10920	0	0	4	112	578	2149	3885	3133	975	83	1	0
	CDH80	4708	0	0	0	18	132	834	1898	1468	340	18	0	0
	WSAvg	4.8	3.8	4.2	5.1	5.7	5.6	5.6	5.5	5.3	5.0	4.1	4.0	4.0
	PrecAvg	30.9	1.4	1.5	2.7	4.4	4.9	3.5	3.3	1.9	1.3	2.1	2.2	1.7
Precipitation	PrecMax	40.7	2.2	2.0	5.0	7.8	7.9	5.5	8.0	4.2	2.5	4.8	5.4	3.3
	PrecMin	19.2	0.6	0.8	1.3	0.7	1.3	1.5	1.2	0.1	0.3	0.6	0.9	0.7
	PrecStd	5.5	0.5	0.4	0.9	2.0	1.7	1.2	1.7	0.9	0.7	1.3	1.2	0.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	45.0	52.8	71.9	82.8	87.9	94.7	99.0	98.3	91.6	82.4	68.7	50.2
		MCWB	37.7	43.1	52.4	58.3	62.7	64.4	64.3	63.8	60.0	55.8	51.6	42.0
	2%	DB	39.4	46.3	66.4	77.2	83.8	91.1	95.7	95.1	87.9	75.2	60.9	43.6
		MCWB	34.8	39.9	50.4	56.9	61.7	63.7	64.3	63.4	60.3	54.4	49.3	38.2
	5%	DB	35.9	42.5	61.1	73.3	80.4	88.0	93.2	91.7	84.1	70.0	55.1	39.4
		MCWB	32.6	37.2	48.5	55.1	60.1	63.2	64.3	62.8	58.3	52.5	45.8	35.7
	10%	DB	33.6	37.8	56.8	69.4	77.0	85.1	89.8	88.1	80.2	66.0	50.0	36.1
		MCWB	31.1	34.1	46.9	54.1	58.8	62.8	63.7	61.7	57.2	50.6	43.1	33.2
	0.4%	WB	38.6	44.2	55.0	62.8	66.6	69.0	71.4	68.8	64.2	58.0	54.4	42.3
		MCDB	42.6	50.3	67.9	76.3	81.5	84.8	88.2	86.6	84.2	73.8	66.0	46.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	35.4	40.1	52.1	59.4	64.0	67.0	68.3	66.0	61.2	55.7	50.1	38.8
		MCDB	38.4	45.3	63.6	71.2	78.4	83.8	86.0	86.1	80.8	71.0	59.4	42.7
	5%	WB	33.1	37.3	49.3	57.0	62.0	65.5	66.6	64.8	59.7	53.9	46.3	36.0
		MCDB	35.3	41.7	59.2	68.5	76.0	82.5	85.3	84.5	79.3	65.8	52.8	39.0
	10%	WB	30.9	34.6	47.2	55.2	60.1	64.2	65.4	63.5	58.1	52.1	43.5	33.5
		MCDB	33.0	37.3	56.1	67.0	73.4	80.4	84.1	82.8	76.8	63.3	49.1	35.8
Mean Daily Temperature Range	5% DB	MDBR	18.0	16.8	20.5	22.1	23.8	24.9	26.6	27.1	26.7	23.8	18.0	15.9
		MCDBR	20.9	21.3	29.0	30.2	28.9	29.4	30.6	30.8	32.2	30.8	26.3	20.5
		MCWBR	16.0	15.1	15.7	14.5	13.3	11.5	10.4	10.6	11.8	13.5	15.3	15.2
	5% WB	MCDBR	18.4	19.4	25.8	24.7	25.9	25.7	26.8	27.7	29.0	25.8	23.7	19.2
		MCWBR	14.4	14.1	14.4	13.0	13.1	11.4	11.1	10.6	12.1	12.4	14.3	14.4
Clear-Sky Solar Irradiance	taub		0.334	0.380	0.441	0.478	0.452	0.453	0.446	0.417	0.400	0.406	0.373	0.328
	taud		2.147	2.038	1.940	1.911	2.056	2.111	2.155	2.225	2.208	2.086	2.102	2.160
	Ebn at Noon		242	249	248	250	262	261	262	266	260	237	223	231
	Edh at Noon		31	42	52	58	52	49	47	42	40	39	33	28
All-Sky Solar Radiation	RadAvg		638	925	1212	1565	1970	2093	2069	1933	1550	1012	618	494
	RadStd		48	74	94	134	132	85	124	95	83	89	60	41

Historical Trends

		Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (5 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	-365	-454	+250	+130

Nomenclature: See separate page